

Lesson 12 · Google AI Development Kit (ADK)

PDF: [this lesson](#) · **Install (Linux · macOS · Windows):** [guide](#) · [PDF](#)

Part of [local-ai-lab](#) - a hands-on course for building local AI.

Course home: <https://nikolareljin.github.io/local-ai-lab/> **Source:** <https://github.com/nikolareljin/local-ai-lab>

Lessons: [1 · RAG](#) → [2 · MCP](#) → [3 · Hybrid retrieval](#) → [4 · RAG safety](#) → [5 · RAG evaluation](#) → [6 · Repo assistant](#) → [7 · LangChain](#) → [8 · LangGraph](#) → [9 · Ollama tools](#) → [10 · Semantic Kernel](#) → [11 · Bedrock Agents](#) → **12 · Google ADK (you are here)** → [13 · AI-assisted testing](#) → [14 · AI code review](#) → [15 · Docs from changes](#)

Status: planned. Outline below; full published slideshow lesson + step-by-step coming later. ADK **runs locally** (`adk run` / `adk web`) against Gemini. Star the repo to follow along.

The idea

Google's Agent Development Kit (ADK) is an open-source framework for building, evaluating, and deploying agents, designed around **Gemini** but model-agnostic. In this final lesson you rebuild the document agent with ADK and run it locally - closing the loop on "the same agent, six different ways."

What you'll learn

- **ADK agents** - defining an Agent with instructions, a model, and tools
- **Tools** - exposing a Python function (your `search_docs`) as an ADK tool
- **Sessions & runners** - running the agent locally with `adk run` (CLI) or `adk web` (dev UI)
- **Gemini integration** - using `gemini-2.5-flash`, plus how to point ADK at other models
- **Evaluation** - ADK's built-in eval harness for checking agent behavior

The design we'll build

```
from google.adk.agents import Agent
from localrag.config import load_config
from localrag.engine import get_retriever

def search_docs(query: str) -> str:
    """Search the user's local documents and return cited passages."""
    hits = get_retriever(load_config()).search(query, 5)
    return "\n\n".join(f"[{h['source']}: {h['page_number']}] {h['text']}" for h in hits)

root_agent = Agent(
    name="docs_agent",
    model="gemini-2.5-flash",
    instruction="Answer from search_docs results and cite [source:page]. "
    "If it's not in the documents, say so.",
    tools=[search_docs],
)
```

```
# Run locally:  adk run .    (or)    adk web
```

The finale: the `search_docs` function is, once again, Lesson 1's retriever. Across MCP, Ollama, Semantic Kernel, Bedrock, and ADK, the **capability** never changed - only the framework wrapping it. That's the whole point of the course: master the primitives, and every framework is just syntax.

Builds on

Concept	From
<code>search_docs</code> retriever tool	Lesson 1
function/tool calling	Lesson 9
agent frameworks compared	Lessons 7, 8, 10
ADK agents, tools, runners, eval	Lesson 12 (this one)

Prerequisites

Python, a Gemini API key (`GEMINI_API_KEY`), and `pip install google-adk`. [Lesson 1](#) provides the retriever the agent calls.

End of the framework tour

You've built the same local-first document agent six ways - from a hand-rolled RAG pipeline up through MCP, local function calling, Semantic Kernel, Bedrock Agents, and ADK - and you understand every layer.

The last three lessons stop building agents and start pointing them at your own work.

Next lesson

[Lesson 13 · AI-Assisted Testing →](#) - generate tests from a change, run them, and let the failure guide the fix.

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